

Large girth can also be used to force a topological K^r minor. We now need some vertices of degree at least $r - 1$ to serve as branch vertices, but if we assume a minimum degree of $r - 1$ to secure these, we can even get by with a girth bound that is independent of r :

Theorem 7.2.7. (Kühn & Osthus 2002)

[7.3.9]

There exists a constant g such that $G \supseteq TK^r$ for every graph G satisfying $\delta(G) \geq r - 1$ and $g(G) \geq g$.

7.3 Hadwiger's conjecture

As we saw in Section 7.2, an average degree of $cr\sqrt{\log r}$ suffices to force an arbitrary graph to have a K^r minor, and an average degree of cr^2 forces it to have a topological K^r minor. If we replace 'average degree' above with 'chromatic number' then, with almost the same constants c , the two assertions remain true: this is because every graph with chromatic number k has a subgraph of average degree at least $k - 1$ (Lemma 5.2.3).

Although both functions above, $cr\sqrt{\log r}$ and cr^2 , are best possible (up to the constant c) for the said implications with 'average degree', the question arises whether they are still best possible with 'chromatic number' – or whether some slower-growing function would do in that case. What lies hidden behind this problem about growth rates is a fundamental question about the nature of the invariant χ : can this invariant have some direct *structural* effect on a graph in terms of forcing concrete substructures, or is its effect no greater than that of the 'unstructural' property of having lots of edges somewhere, which it implies trivially?

Neither for general nor for topological minors is the answer to this question known. For general minors, however, the following conjecture of Hadwiger suggests a positive answer:

Conjecture. (Hadwiger 1943)

The following implication holds for every integer $r > 0$ and every graph G :

$$\chi(G) \geq r \Rightarrow G \succcurlyeq K^r.$$

Hadwiger's conjecture is trivial for $r \leq 2$, easy for $r = 3$ and $r = 4$ (exercises), and equivalent to the four colour theorem for $r = 5$ and $r = 6$. For $r \geq 7$ the conjecture is open, but it is true for line graphs (Exercise 34) and for graphs of large girth (Exercise 32; see also Corollary 7.3.9). Rephrased as $G \succcurlyeq K^{\chi(G)}$, it is true for almost all graphs.³ In general, the conjecture for $r + 1$ implies it for r (exercise).

³ See Chapter 11 for the notion of 'almost all'.

The Hadwiger conjecture for any fixed r is equivalent to the assertion that every graph without a K^r minor has an $(r-1)$ -colouring. In this reformulation, the conjecture raises the question of what the graphs without a K^r minor look like: any sufficiently detailed structural description of those graphs should enable us to decide whether or not they can be $(r-1)$ -coloured.

For $r = 3$, for example, the graphs without a K^r minor are precisely the forests (why?), and these are indeed 2-colourable. For $r = 4$, there is also a simple structural characterization of the graphs without a K^r minor:

[12.6.2]

Proposition 7.3.1. *A graph with at least three vertices is edge-maximal without a K^4 minor if and only if it can be constructed recursively from triangles by pasting⁴ along K^2 s.*

(1.7.3)
(4.4.4)

Proof. Recall first that every IK^4 contains a TK^4 , because $\Delta(K^4) = 3$ (Proposition 1.7.3); the graphs without a K^4 minor thus coincide with those without a topological K^4 minor. The forward implication is left as an easy exercise. To prove Hadwiger's conjecture for $r = 4$ we only need the backward implication, which we prove by induction on $|G|$.

Let G be edge-maximal without a TK^4 . Then $\kappa(G) \leq 2$, as at the start of the proof of Lemma 3.2.2. Hence either G is a triangle, or the assertion follows from the induction hypothesis by Lemma 4.4.4.⁵ $\square$

One of the interesting consequences of Proposition 7.3.1 is that all the edge-maximal graphs without a K^4 minor have the same number of edges, and are thus all 'extremal':

Corollary 7.3.2. *Every edge-maximal graph G without a K^4 minor has $2|G| - 3$ edges.*

Proof. Induction on $|G|$. $\square$

Corollary 7.3.3. *Hadwiger's conjecture holds for $r = 4$.*

Proof. If G arises from G_1 and G_2 by pasting along a complete graph, then $\chi(G) = \max\{\chi(G_1), \chi(G_2)\}$ (see the proof of Proposition 5.5.2). Hence, Proposition 7.3.1 implies by induction on $|G|$ that all edge-maximal (and hence all) graphs without a K^4 minor can be 3-coloured. $\square$

It is also possible to prove Corollary 7.3.3 by a simple direct argument (Exercise 33).

⁴ This was defined formally in Chapter 5.5.

⁵ The proof of this lemma is elementary and can be read independently of the rest of Chapter 4.

By the four colour theorem, Hadwiger's conjecture for $r = 5$ follows from the following structure theorem for the graphs without a K^5 minor, just as it follows from Proposition 7.3.1 for $r = 4$. The proof of Theorem 7.3.4 is similar to that of Proposition 7.3.1, but considerably longer. We therefore state the theorem without proof:

Theorem 7.3.4. (Wagner 1937)

Let G be an edge-maximal graph without a K^5 minor. If $|G| \geq 4$ then G can be constructed recursively, by pasting along triangles and K^2 s, from plane triangulations and copies of the graph W (Fig. 7.3.1).

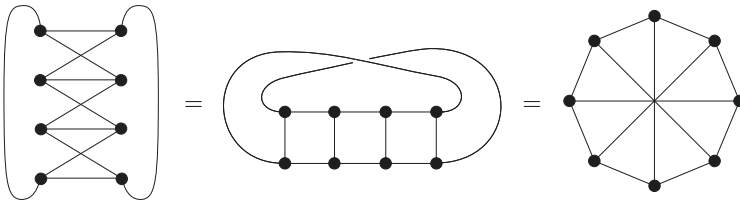


Fig. 7.3.1. Three representations of the Wagner graph W

Using Corollary 4.2.10, one can easily compute which of the graphs constructed as in Theorem 7.3.4 have the most edges. It turns out that these *extremal* graphs without a K^5 minor have no more edges than those that are extremal with respect to $\{IK^5, IK_{3,3}\}$, i.e. the maximal planar graphs:

(4.2.10)

Corollary 7.3.5. A graph with n vertices and no K^5 minor has at most $3n - 6$ edges. $\square$

Since $\chi(W) = 3$, Theorem 7.3.4 and the four colour theorem imply Hadwiger's conjecture for $r = 5$:

Corollary 7.3.6. Hadwiger's conjecture holds for $r = 5$. $\square$

The Hadwiger conjecture for $r = 6$ is again substantially more difficult than the case $r = 5$, and again it relies on the four colour theorem. The proof shows (without using the four colour theorem) that any minimal-order counterexample arises from a planar graph by adding one vertex – so by the four colour theorem it is not a counterexample after all.

Theorem 7.3.7. (Robertson, Seymour & Thomas 1993)

Hadwiger's conjecture holds for $r = 6$.

As mentioned earlier, the challenge posed by Hadwiger's conjecture is to devise a proof technique that makes better use of the assumption of $\chi \geq r$ than just using its consequence of $\delta \geq r - 1$ in a suitable subgraph, which we know cannot force a K^r minor (Theorem 7.2.4). See the notes for some recent advances in this direction.

If we resign ourselves to using just $\delta \geq r - 1$, we can still ask what additional assumptions might help in making this force a K^r minor. Theorem 7.2.7 says that an assumption of large girth has this effect; see also Exercise 32. In fact, a much weaker assumption suffices: for any fixed $s \in \mathbb{N}$ and all large enough d depending only on s , the graphs $G \not\supseteq K_{s,s}$ of average degree at least d can be shown to have K^r minors for r considerably larger than d . For Hadwiger's conjecture, this implies the following:

Theorem 7.3.8. (Kühn & Osthus 2005)

For every integer s there is an integer r_s such that Hadwiger's conjecture holds for all graphs $G \not\supseteq K_{s,s}$ and $r \geq r_s$.

The strengthening of Hadwiger's conjecture that graphs of chromatic number at least r contain K^r as a *topological* minor has become known as *Hajós's conjecture*. It is false in general, but Theorem 7.2.7 implies it for graphs of large girth:

Corollary 7.3.9. *There is a constant g such that all graphs G of girth at least g satisfy the implication $\chi(G) \geq r \Rightarrow G \supseteq TK^r$ for all r .*

^(5.2.3)
^(7.2.7) *Proof.* Let g be the constant from Theorem 7.2.7. If $\chi(G) \geq r$ then, by Lemma 5.2.3, G has a subgraph H of minimum degree $\delta(H) \geq r - 1$. As $g(H) \geq g(G) \geq g$, Theorem 7.2.7 implies that $G \supseteq H \supseteq TK^r$. $\square$

7.4 Szemerédi's regularity lemma

Some 50 years ago, in the course of the proof of a theorem about arithmetic progressions of integers, Szemerédi developed a graph-theoretical tool that has since come to dominate methods in extremal graph theory like none other: his *regularity lemma*. Very roughly, the lemma says that all graphs can be approximated by random graphs in the following sense: every graph can be partitioned, into a bounded number of equal parts, so that most of its edges run between different parts and the edges between any two parts are distributed fairly uniformly – just as we would expect it if they had been generated at random.